Math 871 Topology I
Problem Set 7 solutions

1. The topological space given by $\mathbb{R}$ with the standard topology is a topological field, meaning the following operations are continuous when $\mathbb{R}\times\mathbb{R}$ has the product topology:

$$\begin{array}{cccc} \mu\colon \mathbb{R}\times\mathbb{R}\to\mathbb{R} & s\colon \mathbb{R}\times\mathbb{R}\longrightarrow\mathbb{R} & a\colon \mathbb{R}\longrightarrow\mathbb{R} & i\colon \mathbb{R}\setminus\{0\}\to\mathbb{R} \\ (x,y)\longmapsto xy & (x,y)\longmapsto x+y & x\longmapsto -x & x\longmapsto \frac{1}{x}. \end{array}$$

You may assume all four of these functions are continuous without proof.

(a) Let X be any topological space and $f,g\colon X\longrightarrow\mathbb{R}$ be continuous functions. Show that the following functions are all continuous:

$$\begin{array}{ccc} f\cdot g\colon X\longrightarrow\mathbb{R} & f+g\colon X\longrightarrow\mathbb{R} & -f\colon X\longrightarrow\mathbb{R} \\ x\longmapsto f(x)g(x) & x\longmapsto f(x)+g(x) & x\longmapsto -f(x). \end{array}$$

Moreover, show that if $g(x)\neq 0$ for all $x\in X$, then the following is continuous:

$$\begin{aligned} \tfrac{1}{g}\colon X &\longrightarrow \mathbb{R}\setminus\{0\} \\ x &\longmapsto \tfrac{1}{g(x)}. \end{aligned}$$

Solution. By the universal property of products, the function $\langle f,g\rangle\colon X\longrightarrow\mathbb{R}\times\mathbb{R}$ given by

$$\langle f,g\rangle(x)=(f(x),g(x))$$

is continuous, and it is the unique continuous function such that

$$\pi_1\circ\langle f,g\rangle=f \text{ and } \pi_2\circ\langle f,g\rangle=g.$$

Now note that

$$f\cdot g=\mu\circ\langle f,g\rangle \qquad f+g=s\circ\langle f,g\rangle \qquad -f=a\circ f \qquad \frac{1}{g}=i\circ g$$

are all compositions of continuous functions, and thus continuous.

(b) Show that the following is a closed subset of $\mathbb{R}\times\mathbb{R}$:

$$C=\left\{\left(x,\frac{1}{x}\right)\mid x\in X\right\}.$$

Solution. Since $\mathbb{R}$ is Hausdorff, points are closed, and thus $\{1\}\subseteq\mathbb{R}$ is closed. Since μ is continuous, and

$$C=\mu^{-1}(\{1\})$$

is the preimage of a closed set by a continuous function, it is closed.

2. A function between topological spaces $f\colon X \to Y$ is

$$\begin{array}{ll} \text{an } \textbf{open map} \text{ if} & U \text{ open in } X \implies f(U) \text{ open in } Y \\ \text{a } \textbf{closed map} \text{ if} & C \text{ closed in } X \implies f(C) \text{ closed in } Y. \end{array}$$

(a) Let $f\colon X \longrightarrow Y$ be a function and $\mathscr{B}$ a basis for the topology on X. Show that if $f(U)$ is open for all $U \in \mathscr{B}$, then then f is an open map.

Solution. For any function f and any collection of sets $\{U_\alpha\}_{\alpha\in I}$, we have

$$f\left(\bigcup_\alpha U_\alpha\right) = \bigcup_\alpha f\left(U_\alpha\right).$$

Let U be any open set in X. We can write

$$U = \bigcup_{\alpha\in I} U_\alpha$$

for some collection of open sets $U_\alpha \in \mathscr{B}$. Thus

$$f\left(\bigcup_\alpha U_\alpha\right) = \bigcup_\alpha f\left(U_\alpha\right)$$

is a union of open sets, and thus open.

(b) Prove that if $\{X_\alpha\}_{\alpha\in I}$ is any collection of topological spaces, then under the product topology the projection maps

$$\pi_\beta\colon \prod_{\alpha\in I} X_\alpha \to X_\beta$$

are open maps for all $\beta \in I$.

Solution. By a), we need only to consider a basis for the product topology, so it suffices to consider

$$U = \prod_{\alpha\in I} U_\alpha$$

where U_α is open in X_α for all α and for all but a finite number of α we have $U_\alpha = X_\alpha$. If $U = \emptyset$, then $\pi_\alpha(U) = \emptyset$ for all $\alpha \in I$. If $U \neq \emptyset$, then we claim that $\pi_\alpha(U) = U_\alpha$ for all $\alpha \in I$.
To check this, fix $\beta \in I$ and $u = (u_\alpha)_{\alpha\in I} \in U$. For each $v \in U_\beta$, let

$$x = (x_\alpha)_{\alpha\in I} \qquad \text{where } x_\alpha = \begin{cases} v & \text{if } \alpha = \beta \\ u_\alpha & \text{If } \alpha \neq \beta. \end{cases}$$

Then $\pi_\beta(x) = v$, so $U_\beta \subseteq \pi_\beta(U)$. On the other hand, for any $s = (s_\alpha) \in U$ we have $\pi_\beta(s) = s_\beta \in U_\beta$, and thus $\pi_\beta(U) \subseteq U_\alpha$.
Note: If $U_\alpha \neq \emptyset$ for all $\alpha \in I$, then by the Axiom of Choice the product

$$U = \prod_{\alpha\in I} U_\alpha \neq \emptyset.$$

Equivalently, we can choose $u = (u_\alpha) \in U$. However, note that we did not need the Axiom of Choice to solve this problem.

(c) Give an example showing that projection maps need not be closed.
Hint: You can find a counterexample using $\pi_1 \colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}$.

Solution. Endow $\mathbb{R}$ with the standard topology on all copies, and consider $\pi_1 \colon \mathbb{R} \times \mathbb{R} \longrightarrow \mathbb{R}$. By 1 b), the set

$$C = \left\{ \left(x, \frac{1}{x} \right) \mid x \neq 0 \right\} \subseteq \mathbb{R} \times \mathbb{R}$$

is closed. On the other hand,

$$\pi_1(C) = \{x \in \mathbb{R} \mid x \neq 0\}$$

is not closed, since its closure contains 0: for example, because nonempty open sets are infinite, and thus any open neighborhood of 0 contains points besides 0.

3. Let X be any topological space and $f, g \colon X \longrightarrow \mathbb{R}$ be continuous functions under the standard topology on $\mathbb{R}$.

(a) Prove that $\{x \mid f(x) \geqslant g(x)\}$ is a closed subset of X.

Solution. The subset $[0, \infty)$ of $\mathbb{R}$ is closed, as its complement $(-\infty, 0)$ is open. Moreover, since $f - g$ is continuous and

$$\{x \mid f(x) \geqslant g(x)\} = (f - g)^{-1}([0, \infty))$$

is the preimage of a closed set, this subset is closed.

(b) Prove that the function $h \colon X \longrightarrow \mathbb{R}$ defined by

$$h(x) = \min\{f(x), g(x)\}$$

is continuous.

Solution. Note that $X = A \cup B$, where

$$A = \{x \in X \mid f(x) \geqslant g(x)\} \qquad \text{and} \qquad B = \{x \in X \mid g(x) \geqslant f(x)\}.$$

By (a), both A and B are closed subsets of X. The restriction of a continuous function is continuous, so $g|_A$ is continuous on A and $f|_B$ is continuous on B. Moreover, these two functions agree on

$$A \cap B = \{x \mid f(x) = g(x)\}.$$

By the Pasting Lemma for closed sets, there is a unique continuous function $h \colon X \longrightarrow \mathbb{R}$ such that $h|_A = g|_A$ and $h|_B = f|_B$. By the definition of A and B,

$$h(x) = \min\{f(x), g(x)\} \text{ for all } x \in X.$$

4. Let X and Y be topological spaces and let $f\colon X \longrightarrow Y$ be any function.

   (a) Show that if f is continuous, then f sends convergent sequences to convergent sequences, meaning that for every sequence (x_n) in X converging to x, the sequence $(f(x_n))$ converges to $f(x)$.

   Solution. Let (x_n) be a sequence of points in X that converges to some point $x \in X$. Let U be any open neighborhood of $f(x)$ in Y. Since f is continuous and $f(x) \in U$, $f^{-1}(U)$ is an open neighborhood of x in X. Since (x_n) converges to x, there exists some N such that $x_n \in f^{-1}(U)$ for all $n \geqslant N$. Therefore, $f(x_n) \in U$, for all $n \geqslant N$. This proves that $(f(x_n))$ converges to $f(x)$.

   (b) We say X is **Fréchet-Urysohn** if for every subset $S \subseteq X$ and every $s \in \overline{S}$, there exists a sequence (s_n) of points in S converging to s. Show that if X is Fréchet-Urysohn and f sends convergent sequences to convergent sequences, then f is continuous.

   Solution. Given any closed set C in Y, we will show its preimage $f^{-1}(C)$ is closed. If $f^{-1}(C) = \emptyset$, then it is closed. Suppose $f^{-1}(C) \neq \emptyset$, and let $x \in \overline{f^{-1}(C)}$. Since X is Fréchet-Urysohn, there exists a sequence of points $(x_n) \in f^{-1}(C)$ converging to x. Therefore, $(f(x_n))$ is a sequence in Y converging to $f(x)$. On the other hand, $(f(x_n))$ is a sequence of points in C. If $f(x) \notin C$, then we would be able to find an open neighborhood U of $f(x)$ such that $U \cap C = \emptyset$, and in particular U would contain none of the points in $(f(x_n))$. Therefore, $f(x) \in C$, so $x \in f^{-1}(C)$. We conclude that $f^{-1}(C)$ is closed and thus f is continuous.